

- 4 The leopard, *Panthera pardus*, is a large mammal found in a number of countries in Africa and Asia.

- (a) Most leopards have yellow-brown fur with black spots, but 11% of leopards have black fur.

Figure 4.1 shows the two types of leopard.



Figure 4.1

In the gene that controls fur colour in leopards, the allele that results in black fur is recessive.

Assuming the Hardy-Weinberg principle applies to leopard populations, calculate the percentage of leopards that are heterozygous for the gene that controls fur colour.

Use the equations:

$$p + q = 1$$

$$p^2 + 2pq + q^2 = 1$$

key

p = frequency of the dominant allele

q = frequency of the recessive allele

p^2 = frequency of homozygous dominant genotype

$2pq$ = frequency of heterozygous genotype

q^2 = frequency of homozygous recessive genotype

Show your working **and** give your answer to the nearest whole number.

percentage of heterozygous leopards [3]



- (b) Fur colour in leopards shows discontinuous variation. The mass of leopards shows continuous variation.

Compare the genetic basis of discontinuous variation and continuous variation.

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- (c) Scientists think that the leopard populations in Asia were established by the migration of leopards from Africa approximately 500 000 years ago.

African and Asian leopards are considered to be the same species, but there are many genetic differences between them.

Suggest why there are many genetic differences between African leopards and Asian leopards.

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